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DUM 2026 CONFERENCE SESSION

Distributed Energy Storage in India: Benefits, Challenges, and Key Recommendations for Adoption

Presented By

Bhagyasree Nair | Energy Advisor. GIZ India | Energy Storage for RE Integration in India

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



Project Overview

Energy Storage for Renewable Energy Integration in India (StoREin)

Donor: Federal Ministry for the Environment, Climate Action, Nature Conservation and Nuclear Safety (BMUKN).

Political Partner: Ministry of New and Renewable Energy (MNRE), GoI

12.2024 – 07.2029

Consortium Partners: Fraunhofer IEE, IIT Bombay, TERI, WRI India

giz Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

On behalf of: Federal Ministry for the Environment, Climate Action, Nature Conservation and Nuclear Safety of the Federal Republic of Germany

IKI INTERNATIONAL CLIMATE INITIATIVE

GOVERNMENT OF INDIA MINISTRY OF NEW AND RENEWABLE ENERGY

सत्यमेव जयते

Objective

- Tackling issues linked with intermittent solar and wind power
- Bolstering energy storage capacity, aiming to overcome economic hurdles and knowledge gaps enhancing policy frameworks and demonstrating financial viability
- Utilizing existing and promising future Storage technologies effectively and proposing regulatory recommendations to enhance their viability

Highlights of Results and Impacts

- Framework conditions for the introduction of viable energy storage system applications at the distribution level are improved

Fraunhofer IEE

The Energy and Resources Institute

WRI INDIA

Outputs

III. Technical analysis and R&D networks
- Technical analysis for grid integration
- Establishment of R&D networks for Storage system

I. Policy & Regulation
- Gap analysis
- Recommendations

Energy Storage for Renewable Energy Integration in India

IV. Capacity Building
- Capacity gap analysis
- Training module design
- Training implementation

II. Demonstration projects
- Design and Implementation
- MEL framework (end-of-life plans)

V. Awareness & Dissemination
- Create awareness (Policy makers, research community, DISCOM, training community) on Storage system
- Communication & media



Distributed Energy Storage-A strategic imperative for the modern grids



Turning Distributed Energy Storage Bottlenecks into Growth Pathways

DISTRIBUTION UTILITY MEET DUM 2026

ISGF India Smart Grid Forum

India's clean energy transition requires both utility-scale BESS and DES based solutions, with each option playing a distinct role, and their combined impact ensuring reliability, affordability, and resilience

1. The Rising Role of BESS

BESS: The Backbone of India's Bulk Power System

Generation Side (~70% deployment): Firming RE, reducing curtailment, enabling RTC and FDRE dispatch

Transmission Side: Frequency regulation, voltage support, congestion relief, infrastructure deferral

- ✓ **Current gap:** 120+ GWh tendered (BESS); ~8 GWh deployed (**Less than 1 GWh DES**)
- ✓ **Future requirement:** 236 GWh (BESS); ~**31 GWh (DES) by 2032**

2. Where Bulk BESS Reaches Its Limits

These are not BESS weaknesses; they are simply outside its natural design intent

Technical

- Difficulty in resolving last-mile feeder issues and localized congestion on specific feeders or voltage fluctuations

Operational

- Projects may face issues of grid integration (design, oversizing, etc.), contiguous land availability, regulatory hurdles and high investment costs

Strategic

- Consumer participation/BTM applications needs assets beyond utility-scale assets alone

This creates space where DES adds unique, complementary value

3. Priority applications where DES delivers impact

Smaller, modular storage near consumption/distributed generation

Distribution substation
Peak management, voltage support, feeder congestion relief

Community / microgrid
Shared reliability, rural electrification, islanding support

Rooftop PV/BTM BESS + DES
Self-consumption, energy cost reduction, diesel offset, DSM penalty avoidance, particularly for C&I

EV charging hubs
Demand-charge reduction, renewable based charging of EVs

Critical infrastructure
Hospitals, telecom, data centers etc. for higher RE share at reduced costs

Relevance of DES- a growing imperative for efficient distribution planning

Rising peak demand, increasing RE penetration, & favorable cost economics (ToD front) make DES viable proposition for adoption at state and national level

High Distributed Renewable Penetration

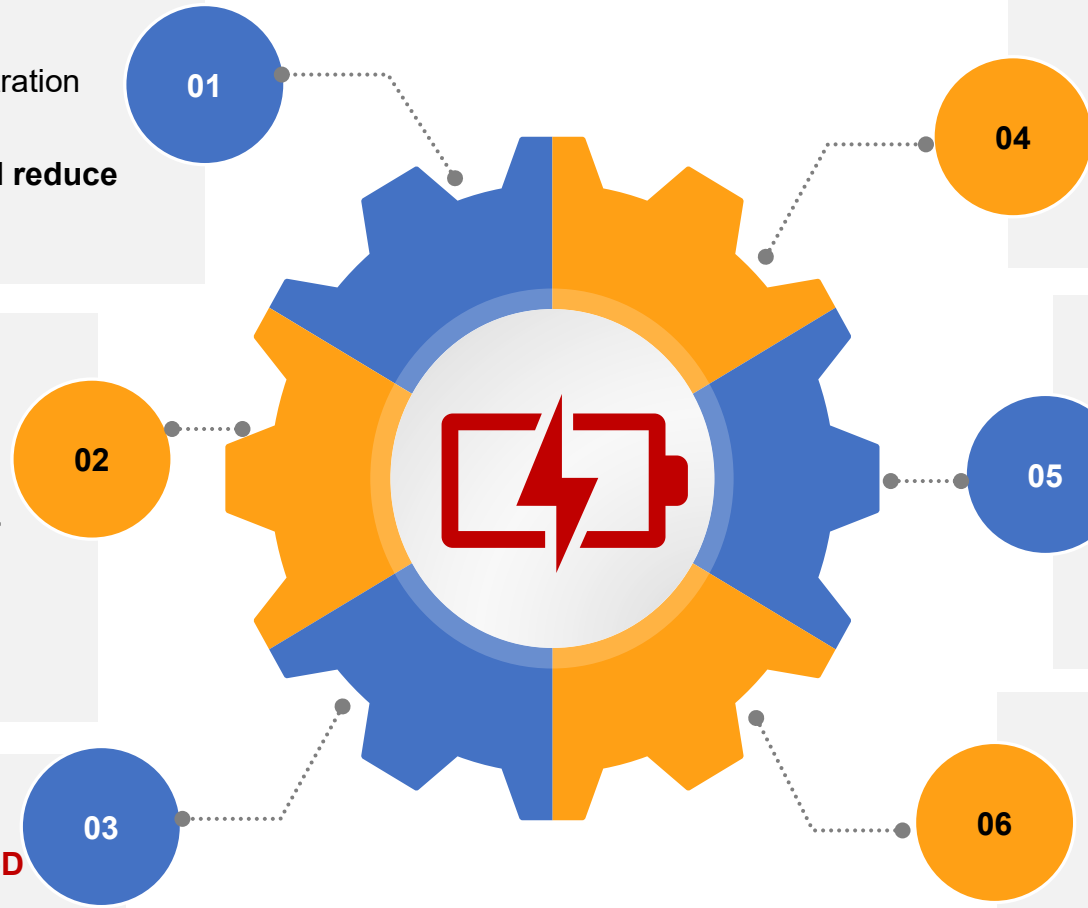
- Growing emphasis on rooftop solar /DRE penetration (PM Surya Ghar & PM KUSUM)
- **DES can help smoothen the variable RE and reduce need for costly T&D upgrades**

Rising peak demand

- India's peak electricity demand has already reached ~270 GW in May 2026. Looking ahead, demand is projected to climb sharply, reaching ~ 366 GW by 2032.
- **DES enables peak shaving and load shifting, avoiding costly emergency purchases by utilities**

Strong economic case

- In several states including **Karnataka, high ToD price arbitrage (~INR 3-4/kWh) makes BTM storage highly profitable, particularly for C&I consumers.**
- **DES cuts costs, reduces diesel reliance and maximizes rooftop solar self-consumption.**



Urban grid stress

- Many states, are increasingly facing DT overloading issues, during peak times
- **DES at feeder or consumer level relieves congestion, deferring expensive upgrades.**

Unreliable power in rural areas

- Multiple states continue to experience outages, with average supply of ~22.5 hours
- **Community-level DES and microgrids can ensure improve energy access in underserved areas.**

DES to be considered while planning

- Multiple state agencies are now integrating storage into their resource adequacy planning.
- **This shows DES can be recognized as essential for balancing renewables and ensuring long-term reliability.**

DES systems carry a structural premium over utility-scale BESS, strengthening the case for stacking multiple value streams

Latest 2026 benchmarks show further decline in utility-scale BESS costs, while distributed systems remain ~1.5–2.1× more expensive due to smaller scale, site works and duplicated balance-of-system costs.

Structural cost gap

a) Utility-scale benchmark

- All-in installed cost for 4-hour+ utility-scale BESS in India is around **USD 100–120/kWh** (Oct 2026 benchmark).

b) Distributed systems

- DES costs vary by site and configuration; use **USD 170–220/kWh** as an illustrative planning range, reflecting smaller scale, site works and grid-integration requirements.

c) Why the gap persists

- Smaller project size, interconnection and site-specific works limit economies of scale.

~110 | **USD/kWh**
All-in utility-scale BESS benchmark (4-hour, India, Oct 2026)

Cost stack drivers

a) Battery prices continue to fall

- Stationary storage battery-pack prices averaged about **USD 60–65/kWh** in 2026 (BNEF); pack prices are not total installed system costs.

b) Balance-of-system costs

- PCS, controls, enclosures, cooling, protection and installation add **~USD 40–60/kWh**, with higher relative share in DES.

c) Site-specific overheads

- Interconnection, civil works, fire safety, metering and SCADA can add **~USD 20–40/kWh**, and weigh more heavily on smaller projects.

~50 | **USD/kWh**
Indicative installation and grid-connection component (India, 2026; varies by site)

LCOS and policy design

a) Higher LCOS for smaller systems

- Lazard 2026 benchmarks (global): Utility-scale standalone BESS **USD 81–182/MWh**; C&I standalone BESS **USD 233–418/MWh**.

b) Value beyond energy shifting

- Network deferral, voltage support, reliability and resilience services can add value but are not fully captured in energy-only LCOS.

c) Policy and market design

- Enable multiple revenue streams, suitable dispatch rules and compensation for local grid services to improve DES viability.

~2.0× | **LCOS midpoint ratio**
C&I vs utility-scale (Lazard 2026; ~326 vs ~132 USD/MWh)

The premium is not a technology failure. It reflects scale, project configuration and local integration costs. Well-designed market and regulatory frameworks that reward multiple grid services can improve DES viability.

Key policy and regulatory developments at national level

GoI has introduced multiple policies and regulatory interventions from demand and supply side perspective to promote deployment of BESS; However, more is needed in DES space

Demand side enablers

- National Framework for Energy Storage in India recommends mandatory **5% energy storage** participation in all RE tenders
- **CEA Technical Standards** mandates BESS ≥50 MW to provide additional services such as AGC, **grid-forming inverter and black start capability**
- **BESS allowed to participate** in Secondary and Tertiary **Ancillary markets** along with participation in HP-DAM segment
- **State utilities are recommended** to incorporate a **minimum of 2-hour co-located BESS** equivalent to 10% of installed solar capacity
- GoI has notified **3 tranches under the VGF scheme for BESS** totaling **43.2 GWh** (>35 GWh tendered out already). Current subsidy is ₹18 lakh/MWh
- **PM Surya Ghar 2.0 to add BESS and link incentives to power generation** (a dedicated subsidy component for residential and small commercial BESS paired with rooftop solar)

Supply side enablers

- **0% duty** on equipment for cell manufacturing; to accelerate **domestic battery manufacturing and deployment**
- GoI plans **ALBM** mandating **local usage of BESS containers**, to support domestic manufacturing and reduce import reliance
- MoP has mandated **20% local content requirement** for BESS projects utilizing VGF
- **Exemption of customs duty for critical minerals** and **CMR incentive scheme** announced to bolster the **domestic mineral refining capacity**
- State governments giving **incentives and subsidies to BESS manufacturers** to set up facilities in states
- **10 GWh Stationary PLI scheme for ACC announced** to catalyze domestic cell manufacturing and reduce import dependence

Best practices on DES policy and regulatory framework adopted by states

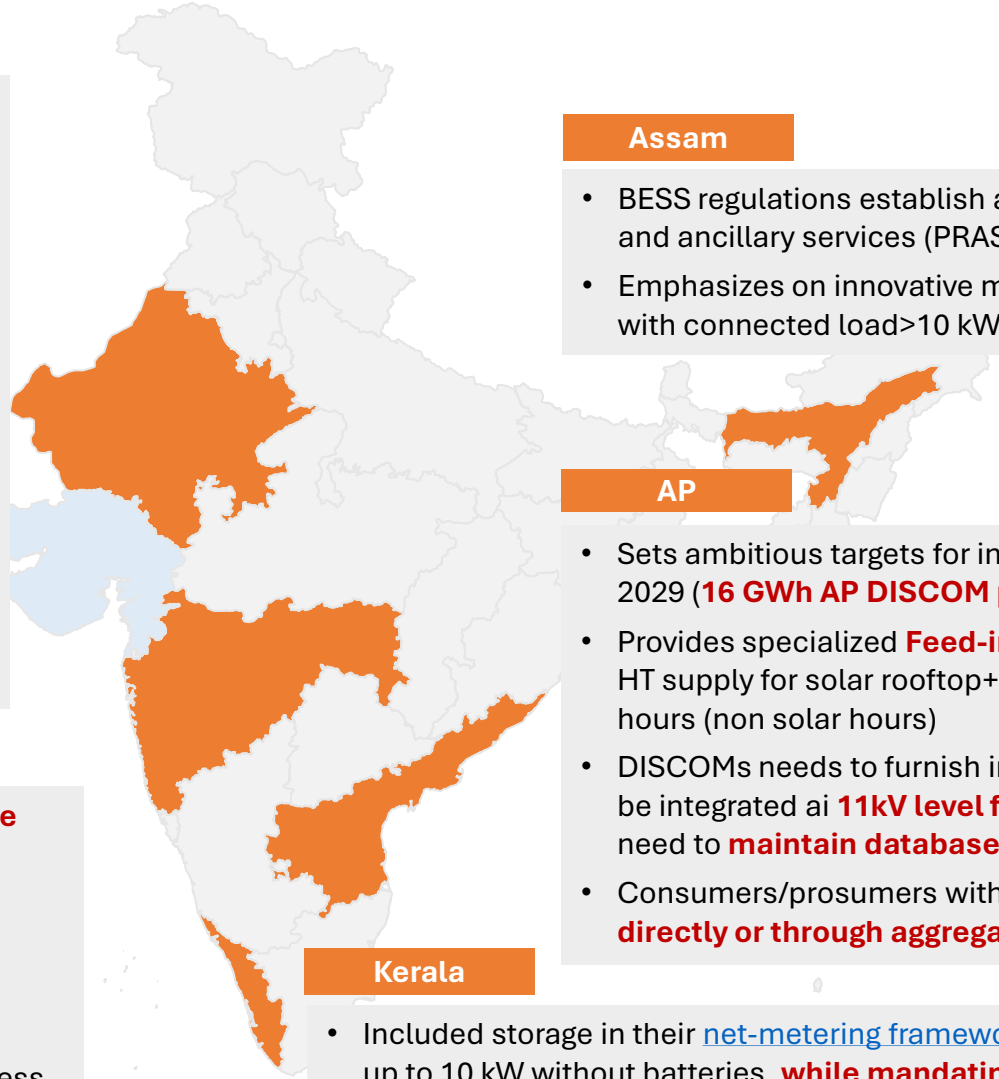
A dedicated state-level BESS policy inclusive of DES, regulatory mandates and enabling net metering framework, and aggregator models can maximize self-consumption, enhance grid flexibility and consumer participation

Rajasthan

- **BESS regulation cover elements of DES** (capacity, duration, etc.), including innovative models for enhancing consumer participation, and aggregation of DERs
- BESS connected at low voltage-11 kV or 33 kV grid sub stations will be **exempted 100% transmission and wheeling charges**
- RE based captive power plants with capacity above 100% of contract demand (up to 200% of contract demand) are **mandated to install BESS for 20%** of the energy generated by additional RE capacity beyond 100% of the contract demand
- Plans to roll out **1 MW / 4 MWh small scale DES pilot project tenders** connected at 33/ 11 kV substations & DT level BESS

Maharashtra

- Any new rooftop solar or open access project **above 100 kW** must integrate **battery storage equal to at least 50% of the RE capacity** for a min. of 2 hours
- Establish an R&D innovation centre to :
  - ✓ Explore pilot initiatives such as 'Storage-as-a-Service' and V2G application
  - ✓ **Assess technical feasibility**, regulatory readiness, and **commercial models suitable for DES**



Assam

- BESS regulations establish a clear framework for tariff determination, dispatch, and ancillary services (PRAS, SRAS, TRAS)
- Emphasizes on innovative models like **P2P trading for residential consumers** with connected load>10 kW, promoting DES

AP

- Sets ambitious targets for integrating **25 GWh of battery storage capacity** by 2029 (**16 GWh AP DISCOM procurement**)
- Provides specialized **Feed-in-tariff of INR 4.17/unit under gross metering** in HT supply for solar rooftop+ BESS projects that can provide power during peak hours (non solar hours)
- DISCOMs needs to furnish information on websites where BESS systems can be integrated ai **11kV level for solarization and SRT installations**. DLs also need to **maintain database for BTM BESS**
- Consumers/prosumers with BTM BESS, are eligible to participate in **AS market directly or through aggregator**

Kerala

- Included storage in their net-metering framework, allowing domestic systems up to 10 kW without batteries, **while mandating 10% storage for systems between 10-15 kW and 20% storage for system between 15-20 kW**

Source: Kerala, AP, Rajasthan, Maharashtra, Assam

Key focus areas for DES growth at national and state level (1/3)

DES in India is being held back not by technology, but by limited policy design

1

Develop a Dedicated National Level DES Policy

- Clearly define DES as a standalone, independently metered asset class (BTM, community level)
- Advance DES storage growth through long term roadmap (focus areas, Y-o-Y targets, resource development needs, etc.), similar to BESS

2

Mandate or Incentivize DES Storage Pairing with Solar Rooftop

- Mandate PV + storage co-location for new RTS above certain threshold (e.g. ≥100 kW for C&I)
- Issuing relevant amendment in state regulations for retrofitting existing DRE/solar projects with DES
- Inclusion of specific incentivized tariff under gross metering for PV+DES, particularly during peak hours

3

Introduce a Standardized Revenue-Stacking Framework to Unlock Multiple Value Streams for DES

- A robust value-stacking model is needed to enable simultaneous earnings across retail savings, ancillary services, demand charge reduction, wholesale arbitrage, power exchanges, and capacity remuneration.

4

Establish a Robust VPP Aggregator and Market Framework for DES

- Lowering the ancillary service threshold from 1 MW would let pooled kW-scale DES; household batteries, small commercial units, EV chargers and qualify for market participation.
- Establish a regulatory framework for BTM pooling and community aggregation, enabling aggregators to combine rooftop solar, storage, and other small-scale assets into VPPs that can participate in wholesale markets, ancillary services, and demand response.

Key focus areas for DES growth at national and state level (2/3)

Scaling DES requires dedicated financial support, expanded Time-of-Day (ToD) regimes, and clear safety standards and operational frameworks

5

Introduce Dedicated Financial Instruments and Incentives to Accelerate DES Adoption

- Create a dedicated national DES financing facility - VGF, capital grants, and low-interest loans to bridge per-unit cost gaps and attract private capital

6

Focus on restructuring ToD regime to Drive DES Adoption

- Widening ToD differential (1.5x-2.0x) for C&I categories, would create significant arbitrage opportunities
- Inclusion of residential consumers/cooperative societies (above certain kW, with smart meters) in the ToD regime (as globally practiced, e.g. Australia)

7

Bridge the Pilot-to-Program Gap by Institutionalizing Scale-Up Pathways for DES

- Boost adoption by launching innovative pilots that demonstrate value stacking principles (e.g., demand charge savings, ToD arbitrage, rooftop solar + BESS), aggregator-based projects, etc.

8

Strengthen DES-Specific Safety Standards and Build Indigenous Testing Infrastructure

- Ensure DES adoption by mandating safety and performance compliance with BIS/ UL)/ IEC standards (IS 17067, IEC 62933, UL9540) and avoiding post commissioning regulatory shifts
- Develop testing facilities (BIS-accredited labs) and skilled manpower to allow indigenous container testing for DES in India (largely BESS containers are being shipped to China for testing and certification)

Key focus areas for DES growth at national and state level (3/3)

Accelerating DES adoption demands a twin-track approach strong national-level data visibility and net metering, alongside dedicated state-level DES policies that recognize DES as NWA and integrate it into DISCOM-level planning

9

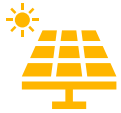
Establish a Centralised National Database for Mapping and Monitoring DES Installations



- Establish DES Data repository tracking all DES/BESS/BTM projects deployed (monthly/yearly)
- Document and share successful case studies with diverse stakeholders through technical performance reports

10

Focus on leveraging high rooftop/DRE penetration with DES (GFM)



- Mandate GFM-capable storage in high RE penetration areas to act as distributed grid-forming substitutes, enhancing local voltage, grid and frequency stability

11

DISCOMS can assess DES as an NWA within distribution planning



- Evaluate DES as a flexible Non-Wires Alternative (flexible grid solution) before conventional upgrades like transformers, feeders, or substations.
- Assess DES against traditional investments on peak load management, T&D loss reduction, and reliability improvement to guide cost-effective grid decisions.

12

Establish planning and procurement for DES projects



- Conduct distribution-level assessments to identify substations/feeders where DES can alleviate DT overloading, pinpoint stress hotspots, and estimate site-specific storage potential.
- Design cost-effective procurement mechanisms (competitive tendering by DISCOMs) and integrate DES into long-term resource adequacy planning.

Case study – South Australia Simply Energy (SE) <> Tesla (Stage 1 – VPP)

Making distributed energy storage bankable through a multi-service VPP model and platform (deX platform)

1200 homes enrolled
5kW PV + 13 kWh battery system each home

AUD 30M loan
and AUD 2 million grant support (early phase)

30% household bill savings
AUD 190 million cumulative system savings (expected)

15.6 MWh battery capacity
6 MW solar; to be later extended to 40,000 homes (>600 MWh capacity)

STAKEHOLDER BUSINESS FLOW: ROLES AND BENEFITS

→ ***Fleet data*** *(battery status, SoC, availability, etc.)*
⇢ ***Dispatch signal*** *(charge, discharge signal etc.)*
→ ***Revenue flow*** *(FCAS, arbitrage, grid service.)*

GOVERNMENT BODIES (ARENA / SA GOVT.)

Roles / service provided
- Provides funding support (grants, loans)
- Set policy framework and incentives
- Derisks early deployment

Benefits realized
- Clean energy transition, grid reliability, & consumer savings

HOUSEHOLDS (HOUSING SA TENANTS)

Roles / service provided
- Host solar rooftop (existing PV system), purchase BESS
- Allow remote control for VPP operations (connects to VPPx)
- Low-income tenants selected

Benefits realized
- Lower electricity bills (>20%), get fixed 2 USD per day or consumption-based rate from SE

TESLA (OEM + VPP OPERATOR)

Roles / service provided
- Supplies, installs asset fleet
- Operates and optimizes VPP platform (partner **GreenSync**)
- Dispatches battery fleet in real-time

Benefits realized
- Platform fees, or service revenue from aggregators

SIMPLY ENERGY (RETAILER / AGGREGATORS)

Roles / service provided
- Developed smart storage offer to onboard eligible customers
- Agg. battery fleet (right to request DER to dispatch)
- Manage settlements
- Share revenue with households

Benefits realized
- Aggregator margin, performance-based incentives

MARKETS AND GRID (AEMO & NETWORK OPERATOR)

Roles / service provided
- AEMO – enable market access (FCAS, etc.) and dispatch
- SA power networks – grid connection, network support

Benefits realized
- System reliability, market efficiency, network stability

Source: ARENA, and PwC analysis

Proven Benefits

- ✓ 20% cheaper tariff for low-income tenants in SA compared to default market operator
- ✓ 1st residential VPP accredited for contingency FCAS in Australia
- ✓ Supports up to 20% of SA's daily energy demand – 8 MW

Learnings for India

- ✓ Standardize VPP-ready Infrastructure (aggregator, VPP operators/software and technology providers)
- ✓ Making residential/C&I consumers integral part of DES programs through carefully designed incentive models

THANK YOU

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